Model Building

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Outline

Intro to Model Building

Choosing variables to include in a regression analysis

When datasets include a large number of variables, it can be a challenge to choose which variables should be included in the regression model and which should not.

Example Problem

Fresh example

Company A produces Fresh, a brand of liquid laundry detergent. In order to manage its inventory more effectively and make revenue projections, this company would like to better predict the demand for Fresh. To develop a prediction model, the company has gathered data concerning demand for Fresh over the last 30 sales periods. The first few lines of this dataset are shown below:

```
> head(fresh)
  demand fresh_price ads_expenditure  size ads_campaign competitor_price
1   7.38        3.85            5.50 Small            B             3.80
2   8.51        3.75            6.75   Big            B             4.00
3   9.52        3.70            7.25   Big            B             4.30
4   7.50        3.70            5.50 Small            A             3.70
5   9.33        3.60            7.00   Big            C             3.85
6   8.28        3.60            6.50 Small            A             3.80
```

- y (demand) = the demand for Fresh (100 000s of bottles).
- x_1 (fresh_price) = the price of Fresh detergent (in 10 rands).
- x_2 (ads_expenditure) = company's advertising expenditure to promote Fresh detergent (in 1000 rands).
- x_3 (size) = size of company (big or small).
- x_4 (ads_campaign) = company's advertising campaign (A: TV commercials, B: mixture of TV & radio, C: mixture of TV, radio, magazine & newspaper ads).
- x_5 (competitor_price) = average industry price of competitor's similar detergents (in 10 rands).

Akaike Information Criterion (AIC)

Akaike Information Criterion (AIC)

- AIC is a criterion that is used to compare different models fitted to the same set of data.
- It assesses the relative quality of models fitted to the same data (ie. it measures how well models fit the data used to generate them)

$$AIC = n \times log\left(\frac{SS_{resid}}{n}\right) + 2p + 2$$

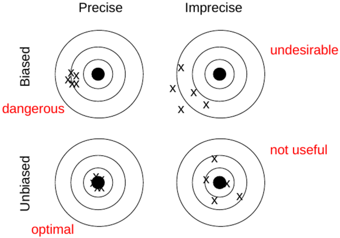
The $log\left(\frac{SS_{resid}}{n}\right)$ is higher when model fit is bad and lower when model fit is good. The $2p$ is higher when more variables are included and lower when fewer are included.

So the AIC is a trade-off between how well a model fits (goodness-of-fit) and how many independent variables are included (complexity).

Using the AIC instead of a goodness-of-fit measure like R^2 is important when comparing different models. This is because R^2 is only a measure of the explained variance in the response and not a trade-off between goodness-of-fit and complexity.

AIC seeks to balance precision and bias

- **Precision** refers to how close the predictions are to actual observations (ie. in our data set, how accurately does our model predict the response).
- **Bias** refers to how much uncertainty there is around the model estimates (ie. each time we fit the regression model to different data, how much does it change).



Including too many independent variables can lead to the model fitting our data set very well (precise) but can lead to the model not fitting unseen data well (biased).

Including too few (or the wrong) independent variables can lead to the the model not fitting our data well (imprecise) and not performing well on unseen data (biased)

Model Building Approaches

Checking all model subsets

One way to choose a model is to fit models with all possible combinations of independent variables.

For a problem with three independent variables this would look like:

- $y \sim 1$ (Intercept model)
    - $y \sim x_1$
    - $y \sim x_2$
    - $y \sim x_3$
  - $y \sim x_1 + x_2$
  - $y \sim x_1 + x_3$
  - $y \sim x_2 + x_3$
- $y \sim x_1 + x_2 + x_3$ (Full model)

With 3 variables this is manageable but still a lot of models. In general, with r possible independent variables, this approach would involve fitting 2^r models. These models then need to be compared using some criterion (like AIC).

For a dataset with 10 variables, this would involve fitting $2^{10} = 1024$ models. This is not practical!

Variable selection procedures

- To avoid having to compare all possible variable combinations, we can use a variable selection procedure to end up with a good model (by AIC standards)
- There are 3 different selection procedures you can use: forward selection, backwards elimination and step-wise selection.
- They all involve iteratively adding and/or removing variables from a starting model and evaluating the effect on the AIC at each iteration.

Forward selection algorithm

1. Start with the intercept model ($y \sim 1$).
2. Add each remaining independent variable to the model separately and calculate the AIC of the resulting model.
3. The independent variable that reduces the AIC the most is added to the model in this iteration.
4. Repeat steps 2 and 3 until there are no further reductions to the AIC (adding no independent variables results in the best AIC value).

Important to note that once a variable is added it cannot be removed again at a later stage.

Forward selection in R

```r
# model with just intercept
fit.intercept <- lm(demand ~ 1, data = fresh)
# full model with all variables
fit.full <- lm(demand ~ ., data = fresh)

# need to specify intercept model as lower and full model as upper and
# use intercept as starting point
step(fit.intercept, direction = "forward",
     scope = list(lower = fit.intercept, upper = fit.full))
```

```
## Start:  AIC=-22.05
## demand ~ 1
##
##                     Df Sum of Sq     RSS     AIC
## + size               1   10.3959  3.0627 -64.457
## + ads_expenditure    1   10.3267  3.1319 -63.787
## + competitor_price   1    7.3888  6.0698 -43.936
## + fresh_price        1    2.9631 10.4954 -27.508
## <none>                           13.4586 -22.047
## + ads_campaign       2    0.3147 13.1439 -18.757
##
## Step:  AIC=-64.46
## demand ~ size
##
##                     Df Sum of Sq    RSS     AIC
## + ads_expenditure    1   1.11491 1.9478 -76.034
## + competitor_price   1   0.74942 2.3133 -70.875
## + ads_campaign       2   0.42225 2.6405 -64.907
## <none>                           3.0627 -64.457
## + fresh_price        1   0.00965 3.0531 -62.551
##
## Step:  AIC=-76.03
## demand ~ size + ads_expenditure
##
##                     Df Sum of Sq    RSS     AIC
## + competitor_price   1   0.43782 1.5100 -81.673
## + ads_campaign       2   0.52045 1.4274 -81.361
## <none>                           1.9478 -76.034
## + fresh_price        1   0.00089 1.9469 -74.048
```

```
## Step:  AIC=-81.67
## demand ~ size + ads_expenditure + competitor_price
##
##                  Df Sum of Sq     RSS     AIC
## + ads_campaign    2   0.55107 0.95894 -91.294
## + fresh_price     1   0.27271 1.23729 -85.648
## <none>                        1.51000 -81.673
##
## Step:  AIC=-91.29
## demand ~ size + ads_expenditure + competitor_price + ads_campaign
##
##                 Df Sum of Sq     RSS      AIC
## + fresh_price    1   0.47438 0.48455 -109.772
## <none>                       0.95894  -91.294
##
## Step:  AIC=-109.77
## demand ~ size + ads_expenditure + competitor_price + ads_campaign +
##     fresh_price

##
## Call:
## lm(formula = demand ~ size + ads_expenditure + competitor_price +
##     ads_campaign + fresh_price, data = fresh)
##
## Coefficients:
##      (Intercept)         sizeSmall   ads_expenditure  competitor_price
##           8.2974           -0.1942            0.4326            1.4689
##    ads_campaignB     ads_campaignC       fresh_price
##           0.2226            0.4148           -2.3103
```

Backwards elimination algorithm

1. Start with the full model ($y \sim x_1 + ... + x_p$).
2. Remove each remaining independent variable from the model separately and calculate the AIC of the resulting model.
3. The independent variable that reduces the AIC the most is removed from the model in this iteration.
4. Repeat steps 2 and 3 until there are no further reductions to the AIC (removing no independent variables results in the best AIC value).

Important to note that once a variable is removed it cannot be added again at a later stage.

Backwards elimination in R

```r
# use full model as starting point
step(fit.full, direction = "backward")
```

```
## Start:  AIC=-109.77
## demand ~ fresh_price + ads_expenditure + size + ads_campaign +
##     competitor_price
##
##                    Df Sum of Sq     RSS      AIC
## <none>                          0.48455 -109.772
## - size              1   0.05747 0.54202 -108.410
## - fresh_price       1   0.47438 0.95894  -91.294
## - ads_expenditure   1   0.53323 1.01778  -89.507
## - ads_campaign      2   0.75274 1.23729  -85.648
## - competitor_price  1   0.93751 1.42206  -79.473
```

```
##
## Call:
## lm(formula = demand ~ fresh_price + ads_expenditure + size +
##     ads_campaign + competitor_price, data = fresh)
##
## Coefficients:
##      (Intercept)       fresh_price   ads_expenditure         sizeSmall
##           8.2974           -2.3103            0.4326           -0.1942
##     ads_campaignB     ads_campaignC  competitor_price
##           0.2226            0.4148            1.4689
```

Step-wise selection algorithm

1. Start with the intercept model ($y \sim 1$).
2. Add each independent variable to the model separately and calculate the AIC of the resulting model.
3. The independent variable that reduces the AIC the most is added to the model in this iteration.
4. Add each remaining independent variable to the model separately and calculate the AIC of the resulting model. Also remove all currently included independent variables and calculate the AIC of the resulting model.
5. The independent variable that reduces the AIC the most is added (if it is not currently included) or removed (if it is currently included) from the model in this iteration.
6. Repeat steps 4 and 5 until there are no further reductions to the AIC (adding or removing no independent variables results in the best AIC value).

We can now remove variables once they have been added and add variables once they have been removed.

Step-wise selection

```r
# also need to specify lower and upper and use intercept as
# starting point
step(fit.intercept, direction = "both",
     scope = list(lower = fit.intercept, upper = fit.full))
```

```
## Start:  AIC=-22.05
## demand ~ 1
##
##                   Df Sum of Sq     RSS     AIC
## + size             1   10.3959  3.0627 -64.457
## + ads_expenditure  1   10.3267  3.1319 -63.787
## + competitor_price 1    7.3888  6.0698 -43.936
## + fresh_price      1    2.9631 10.4954 -27.508
## <none>                         13.4586 -22.047
## + ads_campaign     2    0.3147 13.1439 -18.757
##
## Step:  AIC=-64.46
## demand ~ size
##
##                   Df Sum of Sq     RSS     AIC
## + ads_expenditure  1    1.1149  1.9478 -76.034
## + competitor_price 1    0.7494  2.3133 -70.875
## + ads_campaign     2    0.4223  2.6405 -64.907
## <none>                          3.0627 -64.457
## + fresh_price      1    0.0097  3.0531 -62.551
## - size             1   10.3959 13.4586 -22.047
##
## Step:  AIC=-76.03
## demand ~ size + ads_expenditure
##
##                   Df Sum of Sq    RSS     AIC
## + competitor_price 1   0.43782 1.5100 -81.673
## + ads_campaign     2   0.52045 1.4274 -81.361
## <none>                         1.9478 -76.034
## + fresh_price      1   0.00089 1.9469 -74.048
## - ads_expenditure  1   1.11491 3.0627 -64.457
## - size             1   1.18406 3.1319 -63.787
```

```
## Step:  AIC=-81.67
## demand ~ size + ads_expenditure + competitor_price
##
##                   Df Sum of Sq     RSS     AIC
## + ads_campaign     2   0.55107 0.95894 -91.294
## + fresh_price      1   0.27271 1.23729 -85.648
## <none>                         1.51000 -81.673
## - competitor_price 1   0.43782 1.94783 -76.034
## - size             1   0.67400 2.18401 -72.601
## - ads_expenditure  1   0.80331 2.31332 -70.875
##
## Step:  AIC=-91.29
## demand ~ size + ads_expenditure + competitor_price + ads_campaign
##
##                   Df Sum of Sq     RSS      AIC
## + fresh_price      1   0.47438 0.48455 -109.772
## <none>                         0.95894  -91.294
## - ads_campaign     2   0.55107 1.51000  -81.673
## - competitor_price 1   0.46844 1.42738  -81.361
## - size             1   0.59054 1.54948  -78.898
## - ads_expenditure  1   0.84524 1.80417  -74.333
##
## Step:  AIC=-109.77
## demand ~ size + ads_expenditure + competitor_price + ads_campaign +
##     fresh_price
##
##                   Df Sum of Sq     RSS      AIC
## <none>                         0.48455 -109.772
## - size             1   0.05747 0.54202 -108.410
## - fresh_price      1   0.47438 0.95894  -91.294
## - ads_expenditure  1   0.53323 1.01778  -89.507
## - ads_campaign     2   0.75274 1.23729  -85.648
## - competitor_price 1   0.93751 1.42206  -79.473

##
## Call:
## lm(formula = demand ~ size + ads_expenditure + competitor_price +
##     ads_campaign + fresh_price, data = fresh)
##
## Coefficients:
##      (Intercept)        sizeSmall  ads_expenditure  competitor_price
##           8.2974          -0.1942           0.4326            1.4689
##    ads_campaignB    ads_campaignC      fresh_price
##           0.2226           0.4148          -2.3103
```

Why most data analysts will not rely on these algorithms

- This algorithm does not take into account any practical knowledge about the data (is there correlation between independent variables, does it not make real-world sense to include certain variables, should certain variables be included because they are control variables for example).
- These algorithms also have an inflated chance of capitalising on chance features of our sample. $\longrightarrow$ this means the model won't fit well to new data not used to fit the model.

Instead many data analysts prefer to rather fit models based on extensive exploratory data analysis and good understanding of the data being modelled.